Chapter 2 — Atomic Structure

Exercise Solutions · Full textbook exercise · PECTAA Chemistry 11 · Book pp. 40–42

Q.1 — Multiple Choice Questions (with correct option and reason)

I. The quantum number ‘m’ of a free gaseous atom is associated with:

(a) the effective volume of the orbital
(b) the shape of the orbital
(c) the spatial orientation of the orbital
(d) the energy of the orbital in the absence of a magnetic field

Answer: (c) the spatial orientation of the orbital — The magnetic quantum number m specifies the orientation of an orbital in space (e.g. p_x, p_y, p_z).

II. When 3d subshell is completely filled, the next entering electron goes into:

(a) 4f
(b) 4s
(c) 4p
(d) 4d

Answer: (c) 4p — Filling order is ...4s → 3d → 4p. After $3d^{10}$ the next electron enters 4p.

III. Quantum number values for 2p orbitals are:

(a) n = 2, l = 1
(b) n = 1, l = 2
(c) n = 1, l = 0
(d) n = 2, l = 0

Answer: (a) n = 2, l = 1 — For a p orbital l = 1; the principal quantum number for 2p is n = 2.

IV. An electron having the set of values: n = 4, l = 0, m = 0 and s = +1/2 lies in:

(a) 2s
(b) 3s
(c) 4s
(d) 4p

Answer: (c) 4s — l = 0 means an s orbital; n = 4 gives 4s.

V. The quantum number values (n, l, m) for the fourth electron of $_9$Be atom are:

(a) 1, 0, 0
(b) 2, 0, 0
(c) 2, 1, 0
(d) 1, 1, 1

Answer: (b) 2, 0, 0 — Be (Z = 4): $1s^2\ 2s^2$. The fourth electron is in 2s, so n = 2, l = 0, m = 0.

VI. The correct order of first ionization energies is:

(a) F > He > Mg > N > O
(b) He > F > N > O > Mg
(c) He > O > F > N > Mg
(d) N > F > He > O > Mg

Answer: (b) He > F > N > O > Mg — He is highest (filled $1s^2$). Among the rest: F > N > O > Mg (N > O because of the stable half-filled $2p^3$).

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VII. A p orbital has a characteristic shape with how many lobes?

(a) 1 **(b)** 2
(c) 3 **(d)** 4

Answer: (b) 2 — A p orbital is dumb-bell shaped with two lobes on opposite sides of the nucleus.

VIII. The three p orbitals in a given energy level are oriented:

(a) Along the same axis. **(b)** At 45º to each other.
(c) Mutually perpendicular to each other along the x, y, and z axes. **(d)** In a complex tetrahedral arrangement.

Answer: (c) Mutually perpendicular to each other along the x, y, and z axes. — The three p orbitals (p_x, p_y, p_z) lie along three mutually perpendicular axes.

IX. How many d orbitals are there in a given energy level?

(a) 1 **(b)** 3
(c) 5 **(d)** 7

Answer: (c) 5 — For l = 2, m = −2, −1, 0, +1, +2 → five d orbitals.

X. What information does the principal quantum number (n) gives us about orbitals:

(a) Size **(b)** Shape
(c) Size and shape **(d)** Spin

Answer: (a) Size — n mainly determines the size and energy of the orbital; shape is given by l.

XI. How many unpaired electrons are present in an atom of cobalt?

(a) Two **(b)** Three
(c) Four **(d)** Five

Answer: (b) Three — Co (Z = 27): [Ar] $3d^7$ $4s^2$. In $3d^7$ there are three unpaired electrons (Hund's rule).

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Q.2 — Short Answer Questions

a. There are three orientations of p-orbital due to three values of magnetic quantum number. Justify it.

For a p orbital l = 1, so m can take 2l + 1 = 3 values: −1, 0, +1. Each value corresponds to one spatial orientation, giving p_x, p_y and p_z along three mutually perpendicular axes.

b. 'I₃' of Mg is much bigger than its 'I₂'. Justify.

Mg: $1s^2$ $2s^2$ $2p^6$ $3s^2$. The first two electrons are removed from the 3s valence shell. After Mg^{2+} is formed it has the stable neon core; the third electron must be removed from this inner noble-gas core, which requires far more energy. Hence $I_3 \gg I_2$.

c. Among the elements Li, K, Ca, S and Kr which one has the lowest first ionization energy? Which has the highest first ionization.

Lowest: K (largest atom, outer electron most shielded). Highest: Kr (noble gas with a stable filled octet).

d. Consider the electronic configuration of the potassium atom (atomic number 19). (i) Write the full electronic configuration using s, p, d, f notation. (ii) Explain why 4s is filled before 3d.

(i) K: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1$.

(ii) According to the Aufbau (n + l) rule, 4s has n + l = 4 + 0 = 4, while 3d has 3 + 2 = 5. The subshell with lower (n + l) is lower in energy and fills first, so 4s fills before 3d.

e. (i) An atom of element X has atomic number 17 and mass number 35. Determine protons, neutrons and electrons. (ii) If it forms an ion with charge −1, how many protons, neutrons and electrons are present?

(i) Protons = Z = 17; electrons = 17 (neutral atom); neutrons = A − Z = 35 − 17 = 18.

(ii) For X^-: protons = 17, neutrons = 18 (unchanged), electrons = 17 + 1 = 18.

f. In the ground state of mercury $_{80}$Hg: (i) How many electrons occupy atomic orbitals with n = 3? (ii) How many occupy 4d orbitals? (iii) How many occupy 4p_z? (iv) How many electrons in the valence shell have spin “up” (s = −½)?

Hg: [Xe] $4f^{14}\ 5d^{10}\ 6s^2$.

(i) n = 3 shell = $3s^2\ 3p^6\ 3d^{10}$ = 18 electrons.

(ii) 4d is part of the Xe core and is fully filled: 10 electrons.

(iii) 4p_z is one of the three 4p orbitals; $4p^6$ gives 2 electrons in 4p_z.

(iv) The valence shell is $6s^2$. The two paired electrons have opposite spins; one has s = −½. So the number with s = −½ is 1.

Textbook check (p. 41): spin “up” is normally s = +½, but the question prints s = −½. The count of electrons with the printed spin value (−½) in the $6s^2$ valence shell is 1.

g. The successive ionization energies for an unknown element are I_1 = 896, I_2 = 1752, I_3 = 14,807, I_4 = 17,948 kJ/mol. To which family does it most likely belong?

The large jump occurs between I_2 and I_3, showing that two electrons are removed relatively easily from the valence shell and the third comes from a noble-gas core. The element has two valence electrons and belongs to Group 2, the alkaline earth metals.

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i_1. Consider the ionization energies for aluminium: I_1 = 580, I_2 = 1815, I_3 = 2740, I_4 = 11,600 kJ/mol. (i) Account for the trend. (ii) Explain the large increase from I_3 to I_4. (iii) List the four aluminium ions in order of increasing size.

(i) Successive ionization energies increase ($I_1 < I_2 < I_3 < I$) because each electron is removed from an increasingly positive ion, so the remaining electrons are held more strongly.

(ii) Al^{3+} has the stable neon configuration ($1s^2\ 2s^2\ 2p^6$). The fourth electron must be removed from this inner noble-gas core, causing the large jump from I_3 to I_4.

(iii) Increasing size: $Al^{4+} < Al^{3+} < Al^{2+} < Al^{+}$. Higher positive charge pulls the remaining electrons closer, so the ion is smaller.

The book prints this part as “i)” before part (h); both are answered here in the printed order.

h. (i) State the general order of filling orbitals up to the 4p subshell. (ii) Explain why 4s is filled before 3d according to the Aufbau principle.

(i) 1s → 2s → 2p → 3s → 3p → 4s → 3d → 4p.

(ii) By the (n + l) rule, 4s (n + l = 4) is lower in energy than 3d (n + l = 5), so 4s fills first.

i_2. Draw the orbital box diagram for the valence electrons of a phosphorus atom (atomic number 15), ensuring Hund’s rule and the Pauli Exclusion Principle.

P: [Ne] $3s^2\ 3p^3$. The 3s box contains a pair with opposite spins; the three 3p boxes each contain one unpaired electron with parallel spins (Hund’s rule). No orbital holds more than two electrons, and paired electrons have opposite spins (Pauli).

Phosphorus (Z = 15): [Ne] $3s^2\ 3p^3$



Hund’s rule: three unpaired electrons; Pauli: paired 3s electrons have opposite spins.

Descriptive / Long Questions

Q.3 What are quantum numbers? Describe briefly principal and spin quantum numbers.

Quantum numbers are a set of four numbers that describe the energy, shape, orientation and spin of an electron in an atom. No two electrons in an atom can have the same set of all four quantum numbers (Pauli).

Principal quantum number (n): values 1, 2, 3... It identifies the main shell (K, L, M...), and largely determines the size and energy of the orbital. Larger n means a larger orbital and higher energy. The maximum number of electrons in a shell is

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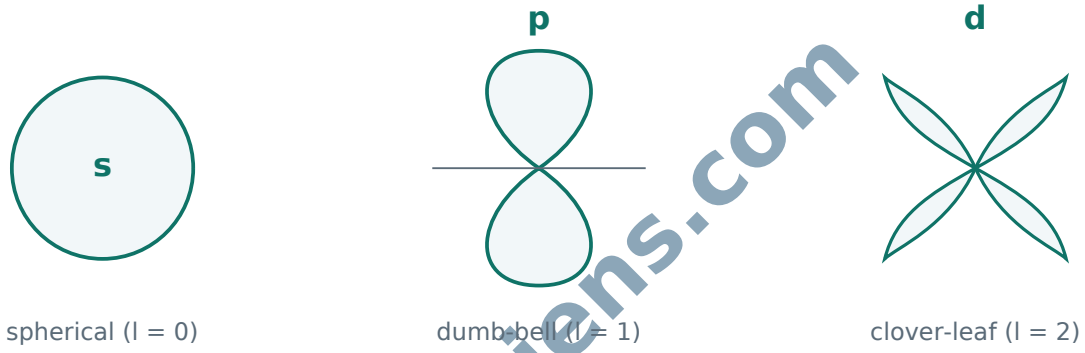
$2n^2$.

Spin quantum number (s or m_s): values +½ and −½. It describes the two possible spin directions of an electron in an orbital. Two electrons occupying the same orbital must have opposite spins.

Q.4 Draw the shapes of s, p and d-orbitals. Justify these by keeping in view the azimuthal and magnetic quantum numbers.

The azimuthal quantum number l determines the shape: l = 0 → s (spherical), l = 1 → p (dumb-bell), l = 2 → d (clover-leaf). The magnetic quantum number m determines the number of orientations: for a given l there are 2l + 1 values of m.

s (l = 0): m = 0 only → one spherical orbital. p (l = 1): m = −1, 0, +1 → three dumb-bell orbitals (p_x, p_y, p_z). d (l = 2): m = −2…+2 → five clover-leaf orbitals.



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Q.5 What do you mean by successive ionization energies? How the electronic shell structure of magnesium (Mg) is derived from the successive ionization energies?

Successive ionization energies are the energies required to remove the first, second, third… electrons one by one from a gaseous atom: I_1, I_2, I_3… They always increase because each electron is removed from a more positive ion.

For Mg the values are approximately I_1 = 738, I_2 = 1451, I_3 = 7733 kJ mol^{-1}. The first two electrons are removed relatively easily, but there is a very large jump at I_3. This shows that Mg has two electrons in its outermost shell; after their removal the stable neon core remains.

Hence the shell structure is 2, 8, 2 ($1s^2$ $2s^2$ $2p^6$ $3s^2$), confirming two valence electrons and placing Mg in Group 2.

Quick Check Solutions (in-chapter practice boxes)

Quick Check 2.1 (book p. 23)

a. Calculate the number of neutrons in (i) $^{39}_{19}K$ (ii) $^{35}_{17}Cl$ (iii) $^{4}_{018}Ar$.

Neutrons = A − Z: (i) 39 − 19 = 20; (ii) 35 − 17 = 18; (iii) 40 − 18 = 22.

b. Which of electron, proton and neutron is deflected the most in a magnetic field?

The electron, because it is charged and has by far the smallest mass, so it experiences the greatest deflection. The neutron is uncharged and is not deflected.

Quick Check 2.2 (book p. 27)

a. Write equations for: (i) 1st I.E. of Ca (ii) 3rd I.E. of K (iii) 2nd I.E. of Li (iv) 5th I.E. of S.

(i) $Ca(g) \rightarrow Ca^{+}(g) + e^{-}$. (ii) $K^{2+}(g) \rightarrow K^{3+}(g) + e^{-}$. (iii) $Li^{+}(g) \rightarrow Li^{2+}(g) + e^{-}$. (iv) $S^{4+}(g) \rightarrow S^{5+}(g) + e^{-}$.

b. For aluminium (Z = 13), sketch $\log_{10}$(successive I.E.) vs number of electrons removed.

The graph rises gradually for the first three electrons ($3p^1$ then $3s^2$), then shows a very large jump at the 4th electron because it is removed from the stable neon core. Further removals rise steeply within the core.

c. From the I_1/I_2 table, identify: (i) reactive metal (ii) reactive non-metal (iii) noble gas (iv) metal forming AX_2.

(i) II (low I_1 = 520, huge I_2 = 7300 → Group 1 reactive metal). (ii) IV (high I_1 = 1680, I_2 = 3380 → reactive non-metal, e.g. F). (iii) I (very high I_1 = 2372, I_2 = 5251 → noble gas, He). (iv) III (I_1 = 900, I_2 = 1760, then a large jump → Group 2 metal forming AX_2, e.g. Be).

Quick Check 2.3 (book p. 30)

a. What information is obtained from (i) principal (ii) azimuthal (iii) magnetic (iv) spin quantum number?

(i) n: shell, size and energy. (ii) l: subshell and orbital shape (s, p, d, f). (iii) m: orientation of the orbital in space. (iv) s: spin direction of the electron (+½ or −½).

b(i). If n = 2 and l = 1, how many orientations are possible?

m = −1, 0, +1 → three orientations ($2p_x$, $2p_y$, 2p_z).

b(ii). If n = 3 and l = 2, which shell and subshell?

Third shell (M) and d subshell (3d).

b(iii). If l = 2, find all possible m values and maximum number of electrons for m.

m = −2, −1, 0, +1, +2. Each m value corresponds to one orbital holding a maximum of 2 electrons; the whole d subshell holds 10 electrons.

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Quick Check 2.4 (book p. 31)

a. What does an orbital represent according to the wave mechanical model?

An orbital is a region of space around the nucleus where the probability of finding an electron is maximum (about 90–95%). It is described by a wave function, not a fixed path.

b. Justify that there are three orientations of p-orbital due to three values of magnetic quantum number.

For p, l = 1, so m = −1, 0, +1 (2l + 1 = 3). Each value gives one spatial orientation: p_x, p_y, p_z along mutually perpendicular axes.

Quick Check 2.5 (book p. 35)

a. Write noble-gas core configurations for (i) $_{48}$Cd (ii) $_{57}$La.

(i) Cd: [Kr] $4d^{10}\ 5s^2$. (ii) La: [Xe] $5d^1\ 6s^2$.

b. Complete electron configuration for antimony (Sb, Z = 51).

$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^6\ 4d^{10}\ 5s^2\ 5p^3$.

c. How many unpaired electrons in $_{51}$Sb?

Three unpaired electrons ($5p^3$, one in each p orbital by Hund's rule).

d. Valence-shell orbital configuration of (i) $_{13}$Al (ii) $_{14}$Si (iii) $_{28}$Ni.

(i) Al: $3s^2\ 3p^1$. (ii) Si: $3s^2\ 3p^2$. (iii) Ni: $3d^8\ 4s^2$.

Quick Check 2.6 (book p. 37)

a. Element with configuration ending $5s^2\ 5p^5$: block, group, period, identity.

p-block; Group 17; Period 5; iodine (I, Z = 53).

b. Configuration $1s^2 \ldots 3d^5\ 4s^1$: block, group, period, name.

d-block; Group 6; Period 4; chromium (Cr, Z = 24). The $3d^5\ 4s^1$ arrangement gives extra stability from a half-filled d subshell.

Quick Check 2.7 (book p. 38)

Write electronic configurations for: (i) Al• (Z=13) (ii) O^{2-} (Z=8) (iii) Fe^{3+} (Z=26) (iv) Cu^{2+} (Z=29) (v) Cu• (Z=29).

(i) Al• (neutral radical): $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^1$. (ii) O^{2-}: $1s^2\ 2s^2\ 2p^6$. (iii) Fe^{3+}: [Ar] $3d^5$. (iv) Cu^{2+}: [Ar] $3d^9$. (v) Cu• (neutral atom): [Ar] $3d^{10}\ 4s^1$.

Textbook checks flagged in this chapter

• p. 41, Q.2(f)(iv): spin "up" is printed with s = −½; conventionally spin up is +½. The answer uses the printed value (−½).

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• p. 41–42: the short-answer lettering is irregular (two parts are printed as “i)” and part (h) appears after the first “i)”). All parts are answered.

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